

Packaging and storage of pharmaceuticals

Learning objective

At the end of this topic students should be able to

- Describe the function of packaging
- Describe the classification of packaging
- Discuss types of packaging material used, their advantage and disadvantage
- Discuss the possible interaction between primary packaging and pharmaceuticals
- Describe the labeling requirement of a packaging system
- The storage conditions required for stability of pharmaceuticals

Introduction

- Packaging is defined as the collection of different components which surround the pharmaceutical product from the time of production to its use.

Functions of packaging

■ Containment

- ✓ Not to leak, nor allow diffusion and permeation
- ✓ Strong enough to hold the contents during handling

■ Protection

- ✓ light
- ✓ moisture
- ✓ oxygen
- ✓ microbial contamination
- ✓ mechanical damage
- ✓ adulteration

Functions of packaging

- **Information**

- ✓ Labels affixed on containers provide information about the product such as composition, appropriate use, storage conditions, etc.
- ✓ Identifies the product from competitors products.

- **Way of administering the drug**

Terms of containers

- **Single-dose containers** hold the medicine that is intended for single use. e.g. glass ampoule.
- **Multidose containers** hold a quantity of the material that will be used as two or more doses. e.g. multiple dose vial or the plastic tablet bottle.
- **Single dose containers multidose containers**



Terms of containers.....

Well-closed containers protect the product from contamination with unwanted foreign materials and from loss of contents during use.

Airtight containers are impermeable to solids, liquids and gases during normal storage and use. If the container is to be opened on more than one occasion it must remain airtight after re-closure.



Terms of containers.....

Light-resistant containers protect the contents from the effect of radiation on the drug.

Amber glass



Terms of containers

Tamper-evident containers are closed containers fitted with a device that irreversibly indicates if the container has been opened.



Child-resistant containers, commonly referred to as CRCs, are designed to prevent children accessing the potentially hazardous product.



Primary and secondary packaging

- **Primary packaging materials** are in direct contact with the product. This also applies to the **closure** which is also part of primary package.
- **Secondary packages** are additional packaging materials that aid transportation and handling of the product.
- **Secondary packages include cartons and boxes** that do not make direct contact with the product.

Dosage form – container interactions

Possible Interactions between primary packaging materials and the included pharmaceutical product:

- Permeation
- Leaching
- Sorption
- Chemical reaction
- Alteration on the properties of the container
- The influence of the manufacturing process (e.g. sterilization) on the container.
- The absorption or adsorption of pharmaceutical components by the packaging materials

Permeation

- The transmission of gases, vapors or liquids from the surrounding environment into the container is known as “*Permeation*”.
- Permeation of water vapor & oxygen through the container wall into the dosage form can be problematic if the drug is sensitive to **hydrolysis and/or oxidation**.
- An increase in temperature, increases permeability of gases.

Leaching

- Release of a constituent **from the container** into the **formulation** is known as “*leaching*”.
- For example, particular dyes which are used as coloring agents may migrate into a product, contaminates the product and may cause a **toxic effect**.

Sorption:

- The Process of extraction / removal of one or more of the constituents **from the formulation** by **the packaging material**.
- Becomes a serious problem particularly for dosage forms that contain drug and/or other important ingredients in the **solution form**.
- May significantly affect the therapeutic efficacy of the formulation containing **highly potent drug**.

Chemical reactivity

- Certain ingredients **used in the container manufacturing** may **chemically react** with one or more **components of a drug product**.
- These chemically incompatible substances may also alter the appearance of the **container or formulation**.

Modification:

- **Is the physical or chemical alteration of the packaging material by the drug product.**
- The content may extract the plasticizer, antioxidant or stabilizer, thus changing the **flexibility** of the container.
- **Permeation, sorption or leaching** may also alter the properties of the plastic container.
- For example: (1) Oils have a softening effect on polyethylene; (2) Fluorinated hydrocarbons attack polyethylene & PVC.

Selection of packaging

The selection of packaging for a pharmaceutical product is dependent on the following factors:

- The nature of the product itself: its chemical activity, sensitivity to moisture and oxygen, compatibility with packaging materials
- The dosage form
- Method of administering the medication
- The type of patient: is it to be used by an elderly patient or by a child.
- Required shelf life

Packaging materials

Glass

- It is the preferred packaging material for many pharmaceutical products.
- **It is composed principally of silicon dioxide**, with varying amount of other oxides such as sodium, potassium, calcium, magnesium, aluminium, boron and iron.



Glass

- The basic structural network of glass is formed by the silicon oxide tetrahedron.
- Boric oxide will enter into this structure, but most other oxides do not. The latter are only loosely bound and are relatively free to migrate.
- The migratory oxides may be leached into a solution in contact with the glass, particularly during the increased reactivity of thermal (heat) sterilization.

Types of glass containers

- The United States Pharmacopeia (USP) classifies glass containers as **Type I, II, III and NP** according to the amount of alkali released from glass **when attacked by (or in intimate contact with water under specified conditions).**

Type I

- Also known as neutral or borosilicate glass.
- Composed principally of silicon dioxide ($\sim 81\%$) and boric oxide ($\sim 13\%$), with low levels of the non-network forming oxides (e.g., sodium and aluminium oxides)
- Possesses a hydrolytic resistance
- The most inert type of pharmaceutical glass (low leach ability) with the lowest coefficient of thermal expansion.
- As a result, it is unlikely to crack on exposure to rapid temperature changes.

Types 1 glass containers

Ampoules



Type I

- Suitable for packaging all pharmaceutical preparations, however, **it is expensive** & this restricts its application.
- Widely used as **glass ampoules & vials to package fluids for injection**. In addition, it is used to package solutions that could dissolve basic oxides in the glass.

Type II (treated soda –lime glass)

- Made of soda-lime-silica glass with a high hydrolytic resistance due to surface treatment of the glass with sulper.
- Soda-lime-silica implies made up of sodium oxide, lime(CaO) and silica.
- Sodium oxide($\sim 14\%$) and calcium oxide ($\sim 8\%$).



Type II (treated soda –lime glass)

- Does not contain boric oxide.
- Type II containers are made of commercial soda-lime glass that has been **de-alkalized or treated to remove surface alkali**.
- The sodium oxide makes the material easy to process and mould while the calcium oxide produces a degree of chemical resistance.

Type III glass

It is a soda-lime glass.

Has more leachable oxides than type 2.

Suitable principally for anhydrous liquids or dry substances (solids).



Type II and III glasses

- Both types melt at lower temperature, are easier to mold into various shapes and have a higher thermal coefficient of expansion. It is thus easier to produce and consequently cheaper.

NP glass type

- It is a soda-lime glass.
- **Type NP release the highest amount of alkali.**
- Must not be used to package parenteral products but it is suitable for packaging solid & semisolid formulations.

USP glass types and uses

Type	General description	General use
I	Highly resistant borosilicate glass	Buffered and unbuffered aqueous solutions. All other uses.
II	Treated soda – lime glass	Buffered aqueous solutions with pH below 7.0, dry powders, oleaginous solutions.
III	Soda-lime glass	Dry powders, oleaginous solutions.
NP	Soda-lime glass	Not for parenterals. For tablets, oral solutions and suspensions, ointments and external liquids.

Advantages of glasses

- ✓ It is inert to most medicinal products.
- ✓ It is impervious to air and moisture.
- ✓ It allows easy inspection of the container contents.
- ✓ It can be colored to protect contents from light.
- ✓ It is easy to clean & sterilize by heat.
- ✓ It is available in variously shaped containers.

The disadvantages of glasses

- ✓ It is fragile.
- ✓ Certain types of glasses release alkali into the container contents.
- ✓ It is expensive when compared to the price of plastic.
- ✓ It is heavy resulting in increased transport costs

Plastics

- Chemically plastics are synthetic polymers of high molecular weight molecules.
- In more recent times, plastic has been developed for the packaging of parenteral products including infusion fluids & small volume injections.

HDPE (High density polyethylene)

- Made from polymerization of ethylene ($\text{CH}_2=\text{CH}_2$) to form $(-\text{CH}_2-\text{CH}_2-)_n$ ($n \geq 1000$)
- In general considered highly protective.
- It is rigid, durable and offer greater chemical resistance
- Has strong intermolecular forces and tensile strength than low-density polyethylene.
- Semi-permeable for liquid preparations.



Types of plastics

LDPE (low density polyethylene)

- it is softer and more flexible than HDPE.
- It has lower melting point and more transparent.



Types of plastics

PP (polypropylene)

- Provides a rigid package with excellent moisture barrier.
- Used primarily **for jars and closures**.
- More resistant to **temperature than HDPE**.



Types of plastics

PVC (Polyvinyl Chloride)

- A commonly used plastic material next to HDPE for pharmaceutical packaging.
- PVC possesses good fabrication flexibility, clarity, and low cost.
- Used for manufacture packaging of IV bags, blister packs, etc.



The principal plastic materials used in pharmaceutical packaging

Plastic polymer	Properties	Uses	Notes
Low-density polyethylene (LDPE)	Soft, flexible and easily stretched.	Squeeze bottles as eye drop bottles.	Disadvantages. of PE (LDPE and HDPE): •Softened by flavoring agent and aromatic oils, •Unsuitable for packaging oxygen sensitive products, •Adsorb antimicrobial preservative agents, •Crack on contact with organic solvents.
High-density polyethylene (HDPE)	Strong, stiff, less permeable to gases than LDPE.	Bottles for solid dosage forms	
Polypropylene	Strong and stiff, good resistance to cracking when flexed	Used for closures with hinges. Used also for tablet containers.	
Polyvinyl chloride (PVC)	Rigid	Laminate (for blisters) and the main constituent of IV bags.	

Advantages of plastics

- ✓ Are flexible & not easily broken
- ✓ Are of low density & thus light in weight
- ✓ Can be heat sealed
- ✓ Are easily moulded into various shapes
- ✓ Are suitable for use as container, closure & as secondary packaging
- ✓ Are cheap

Disadvantages of plastics

- ✓ It is not as chemically inert as type I glass.
- ✓ Some plastics undergo stress cracking & distortion from contact with some chemicals.
- ✓ Plastics are heat sensitive.
- ✓ Substances such as the active drug & preservatives may be taken from the product.
- ✓ It is not as impermeable to gas & water vapor as glass.
- ✓ It may possess an electrostatic charge which will attract particles.
- ✓ Additives in the plastic are easily leached into the product.

Metals

- Metals used for various pharmaceutical packaging include **aluminum** and **stainless steel**.
- Metal containers are used solely for medicinal products of **non-parenteral administration**.
- Metal containers include **tubes, aerosol cans, blisters and strip packs**.

Metals.....

□ General properties

- ✓ Metal is strong, opaque, and impermeable to moisture, gases, light, etc.
 - ✓ Metal is resistant to high and low temperatures.
-
- Metal is an ideal packaging material for pressurized containers.

Blister packs

- These are used for packaging **unit doses of tablets** and **capsules** and can act as an aid for patient compliance.
- The medication is placed in a compartment in a **base material made of plastic such as PVC** lined on the top with aluminum.



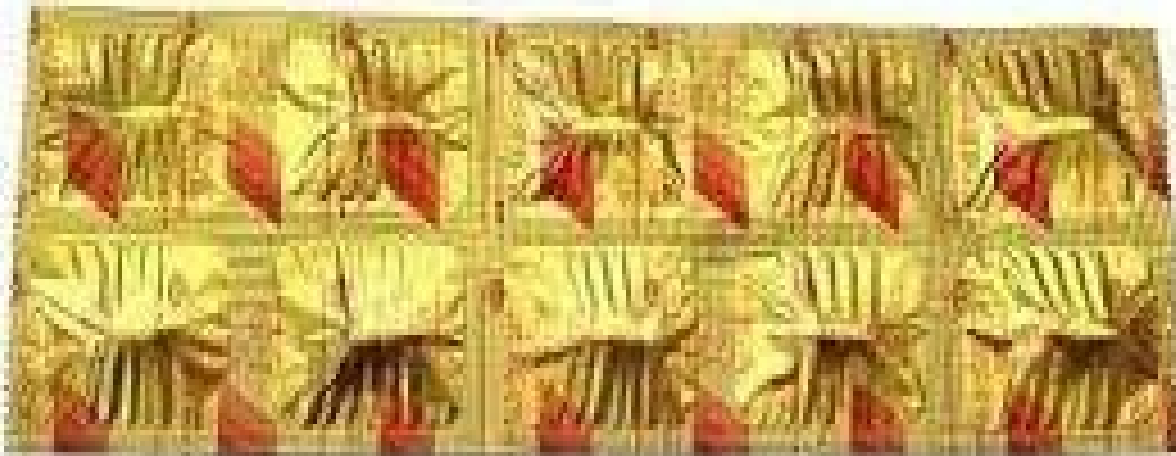
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Blister packs

- Perforations in this material allow individual sections of the package to be broken off.
- **Blister packs are rigid unlike strip packs that are flexible.**

Strip packaging

- With strip packaging, two webs of material sandwich various types of dosage forms such as **tablets, capsules, suppositories, and pessaries**.
- Each of these dosage forms is contained within its own compartment.
- Aluminum foil is commonly used to manufacture strip packs.



Closures

Closures generally require consideration of the following factors:

- **Resistant & compatible** with the product & the main container.
- If closure is re-closable, it should be **readily open able & effectively resealed**.
- Capable of high-speed application for automatic production by high speed machines without loss of seal efficiency.
- Decorative & of a shape which blends in with the main container.
- **Offers such additional functions: aid-pouring, metering, administration, child resistance, tamper evidence, etc.**

Closures

- Prevents exchange with the outside atmosphere to a permissible level.
- Any closure system should provide an **effective seal** to retain the container contents & exclude external contaminants.
- **Child-Resistant Closures (CRCs)** commonly consist of a glass or plastic vial or bottle with a specially designed closure.
- The CRCs in most common use with dispensed medicines is the **push down & turn**.

Tamper-evident closures

- In recent years greater awareness of the vulnerability of products has led to the development of **tamper-evident closures**.
- The closures indicate if **unlawful access** to the container contents has occurred.
- Tamper-evident closures are available in various designs suitable for different containers.

Labeling pharmaceutical dosage forms

- The label on dispensed medicines has two main functions:
 - ✓ to uniquely identify the contents of the container.
 - ✓ to ensure that patients have clear & concise information which will enable them to take or use their medicine in the appropriate way.

Labeling.....

- The label of a pharmaceutical product must be **in the right place and contain the right information.**
- The following need to be taken into consideration:

Appearance

- Correct position
- Clean
- Secure

Labeling.....

❑ Information should be:

- Legible
- Concise
- Adequate
- Intelligible
- Accurate

Types of Information required in pharmaceutical label

INSTITUTION NAME_____	DATE_____
PATIENT NAME_____	
GENERIC NAME, STRENGTH & D _F s OF MEDICINE_____	
FREQUENCY & DURATION OF ADMINISTRATION OF THE MEDICINE_____	
QUANTITY OF THE MEDICINE DISPENSED_____	
HOW TO TAKE THE MEDICINE_____	
EXPIRY DATE_____	
STORAGE CONDITION _____	

Stability of Pharmaceuticals

- Medicines do not keep indefinitely. Some can be kept for only a short time.
- There are many reasons why this is the case.
- These are:-
 - loss of drug (such as hydrolysis or oxidation)
 - loss of vehicle (such as evaporation of water or other volatile ingredients)

Stability.....

- loss of uniformity (such as caking of a suspension or creaming of an emulsion).
- change in bioavailability (particularly with tablets where ageing can affect drug dissolution).
- appearance of toxic products (as a result of chemical change).
- change of appearance (such as color changes).

Storage

- There is a need to specify **storage conditions and shelf life** to ensure effectiveness of the product.
- Medicines should be stored as stated by the manufacturer.
- Storage conditions:
 - ✓ avoid excessive temperature (not above 30 °C for tablets), at about 25 °C for liquids and semi-solids.

Storage conditions

- ✓ protection from light,
- ✓ Storing in dry place (i.e., avoid moisture)
- ✓ Some medicines should be stored in refrigerator (2-8°C) (e.g., vaccines)